

APR-17/BE/Insem.-7

B.E. (Civil Engineering) (Semester - II)

FINITE ELEMENT METHOD IN CIVIL ENGINEERING
(2012 Pattern) (Elective - III)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer Q.1 or 2, Q.3 or 4, Q.5 or 6.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Use of electronic pocket calculator is allowed.
- 5) Assume suitable data if necessary.

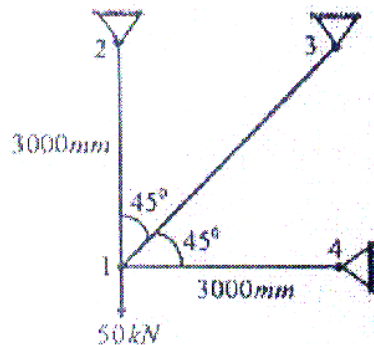
Q1) Derive the saint venant's strain compatibility conditions for 3D elasticity problem. [10]

OR

Q2) Write short notes on plane stress 2D elasticity problem. [10]

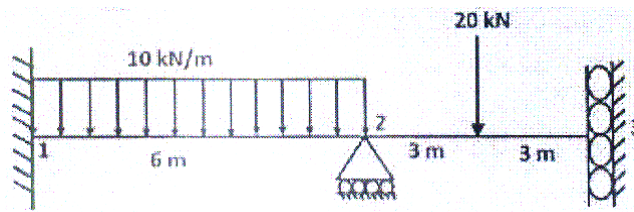
- a) State of a stress at a point
- b) State of strain at a point
- c) Generalized Hooke's law

Q3) For the plane truss composed of three elements shown in figure subjected to a downward force of 50 kN at node 1, determine the displacements at node 1 using finite element method. Take $E = 200 \text{ GPa}$ and $A = 1000 \text{ mm}^2$ for all elements. [10]

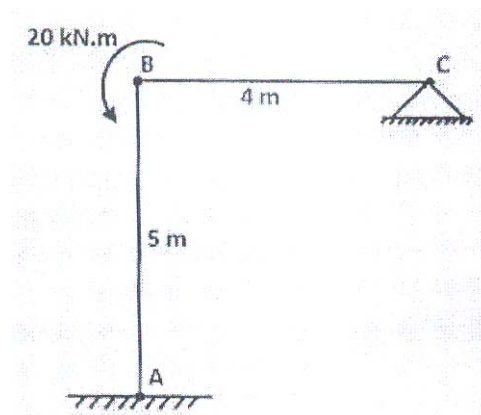


OR

- Q4)** A beam has fixed support at node 1, roller support at node 2 and guided support at node 3. Determine the bending moment diagram and the rotations of nodes 2 and 3 using finite element method. Take $EI = 20 \times 10^3 \text{ kN.m}^2$. [10]



- Q5)** Determine the rotations of joint B and C of the frame as shown in figure using finite element method. Take $EI = 10 \times 10^3 \text{ KN.m}^2$. Neglect axial deformations. [10]



OR

- Q6)** Determine unknown joint displacements at node B of the orthogonal grid as shown in figure. Take $EI = 1500 \text{ kN.m}^2$ and $GJ = 700 \text{ kN.m}^2$ [10]

